MECHANICAL ENGINEERING PORTFOLIO

Avneet Aujla

Mechanical Engineering | University of Guelph

Graduation: April 2028

Manufacturing | Process | Automation | Test

Engineering internships

Internship availability: Summer 2027 or Fall 2027

Mechanical design, manufacturing and automation work spanning press inspection, robot teaching, product transfer and mechanical power transmission.

TOOLS AND ENGINEERING METHODS

SolidWorks | AutoCAD | Cognex IS8912 | TOYOPUC PLC | HMI integration
PLA 3D printing | Tolerancing | Material selection | Fabrication coordination

TOYOTA / SPLIT DETECTION

100%

Split detection; 0 false positives across 15 defective and 15 good test panels.

VISION / PAGES 02-03

TOYOTA / ROBOT TEACHING

$250 vs. $9,600

Filament cost for 32 holders vs. quoted fabrication cost.

FIXTURE / PAGE 04

CARGILL / PRODUCT TRANSFER

95%

Reduction in production-line backups; transfer handled 500+ pieces/hr.

CONVEYOR / PAGE 05

GUELPH / DUAL-BELT MECHANISM

1 motor | 2 belts

Both belts driven from one motor through one chain stage.

MECHANISM / PAGES 06-07

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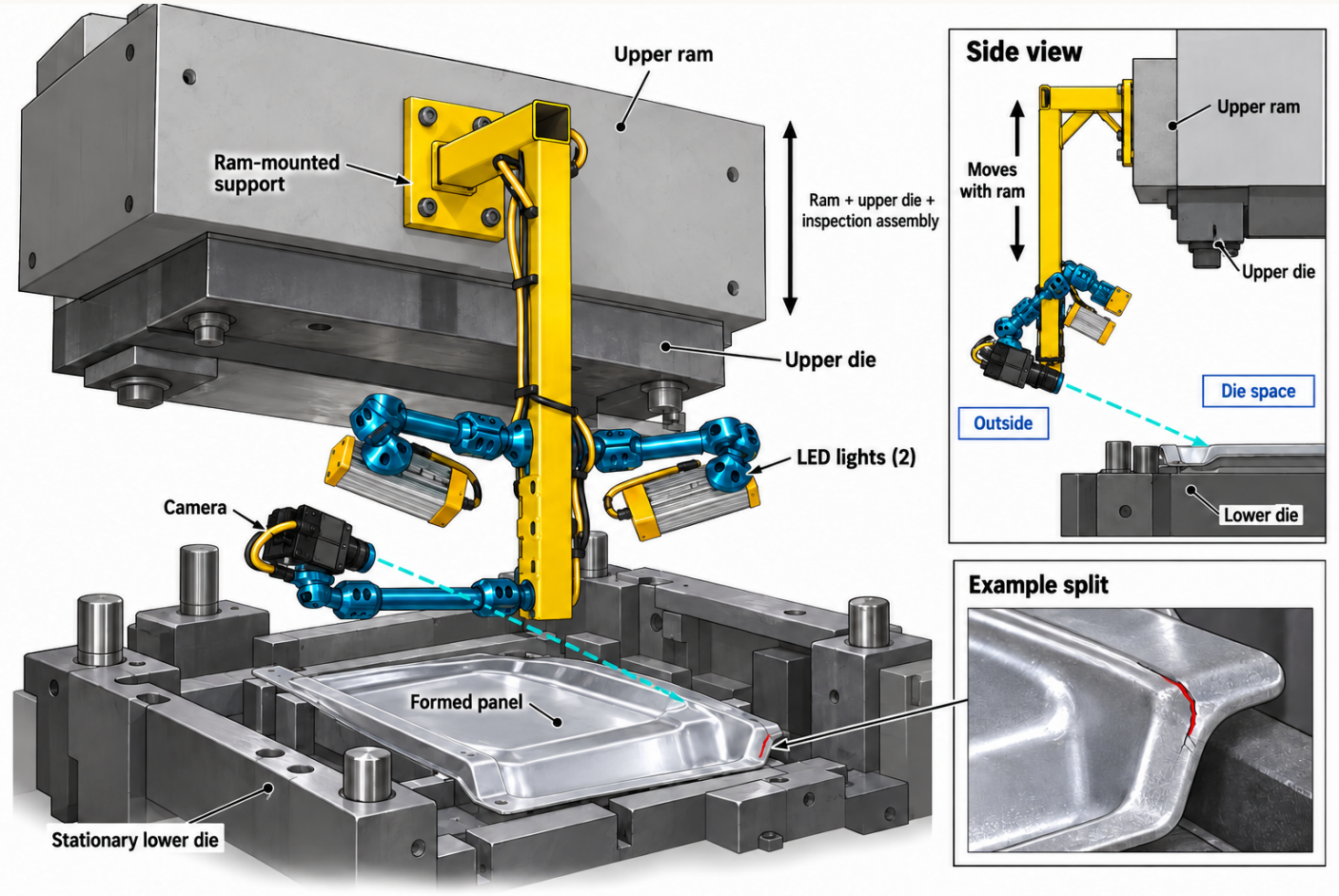
https://avneetaujla.github.io/engineering-portfolio-/

01 / TOYOTA MOTOR MANUFACTURING CANADA

Draw-press split detection

Press Engineering Analyst Co-op | 8 months | January to August 2026
Cognex IS8912, TOYOPUC PLC and HMI integration

Two earlier panel tear-offs left material in the dies, causing **more than 35 repair hours and three missed production runs**. The inspection targets split panels before the next press stroke.



Reconstructed inspection layout: exterior ram mounting, inward view toward the panel, and motion with the upper die.

CAMERA AND LIGHTING ARRANGEMENT

The Cognex IS8912 and two LED lights are mounted on a ram support, outside the die space. They travel with the ram and upper die; the lower die remains stationary.

TRIGGER TIMING AND OPERATOR CONTROLS

Press/robot PLC communication triggers inspection at the robot cup-alignment angle. I specified the PLC trigger logic, signal sources and setup; maintenance implemented the code. I added HMI enable/disable controls and remote camera access.

01 / TOYOTA MOTOR MANUFACTURING CANADA

100% split detection in panel trials

Press Engineering Analyst Co-op | 8 months | January to August 2026
Training, test results and integration handover



Formed panel with marked split and tear-off regions: the defect type targeted by the inspection.

MODEL TRAINING

20 good + 20 defective

The edge-learning model was trained using 20 good and 20 defective images.

PANEL TRIALS

30 test panels

15 defective + 15 good. Panels were separate from the training images; splits varied in size and location.

100%	0
Split detection in the test set	False positives in the test set

Results apply to the tested panels. Training images and panel trials are reported separately.

LIGHTING ADJUSTMENT

Ambient lighting affected detection, including false negatives. Dedicated lighting was added to compensate.

INTEGRATED INSPECTION PATH

PLC trigger
Robot cup-alignment angle

→

Camera + lighting
Cognex IS8912 + two LED lights

→

Split inspection
Trained edge-learning model

OPERATING AT HANDOVER

Camera, lighting, PLC trigger, trained model, HMI enable/disable and remote access were operating at handover.

INTEGRATION PLAN COMPLETED

I completed the automatic PLC cycle-stop integration plan, with documented steps ready for implementation after handover.

02 / TOYOTA MOTOR MANUFACTURING CANADA

An offline fixture for two plants

Press Engineering Analyst Co-op | 8 months | January to August 2026
SolidWorks, PLA printing and tolerancing

I designed the layout, guides, holders and mounts for offline teaching without occupying production equipment. I led the design and coordinated with the maintenance group lead for approval. PLA was adequate for the holder loads in this offline application.

$250 filament vs. $9,600 fabrication quote

32 holders | Filament cost: 97.4% below quote

CAMBRIDGE CONFIGURATION

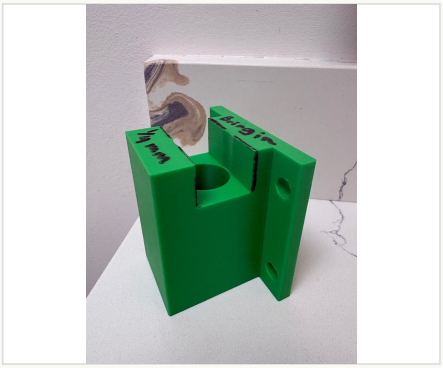
Holders at conveyor fronts, sides and side guides.

WOODSTOCK CONFIGURATION

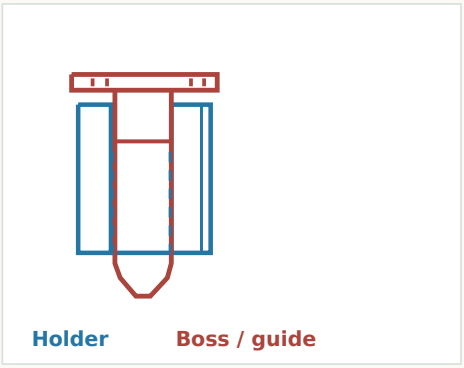
Holders on both sides of each part table.



Print 1: too tight.



Print 2: slightly loose.



Holder and removable boss/guide interface; schematic.

FIXED HOLDERS, REMOVABLE GUIDES

The table and holders stay mounted; bosses and guides are swapped. The three-print iteration reached an accepted sliding fit installed at Toyota.

PRINT	OBSERVED FIT	CHANGE / OUTCOME
1	Too tight	Increased hole diameter and slot width; print 2 overshot.
2	Slightly loose	Reduced hole diameter and slot width from print 2 for print 3.
3	Accepted sliding fit	Installed at Toyota.

Comparison covers the 32 holders: quoted fabrication versus filament for in-house printing, not a total manufacturing-cost comparison.

03 / CARGILL

95% fewer production-line backups

Project Engineering Intern | 4 months | April to August 2025
AutoCAD layout, contractor design reviews and fabrication coordination

500+ pieces/hr
Handled through the transfer

95%
Reduction in production-line backups

ENGINEERING SCOPE & PRODUCT ROUTE

I developed the AutoCAD layout, led contractor design reviews and coordinated fabrication. The transfer redirects raw meat accumulating on Line 3 into Line 4's greater packaging capacity, while retaining Line 3's existing route.

DESIGN DECISIONS

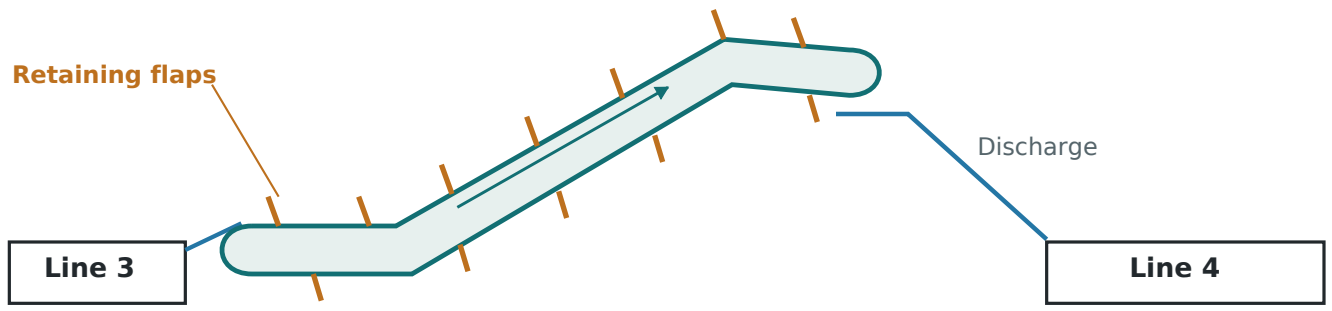
DECISION	ENGINEERING BASIS / CHANGE
Incline layout	Set the incline to clear existing equipment and keep employee work areas below unobstructed.
Retaining flaps	Flap geometry set for rollback prevention on the incline.
Motor / drive	Selected a single-speed plant motor to match the other conveyors, for reliability and to avoid unnecessary complexity.
Contractor review	Layout adjusted to fit existing equipment.

CALCULATION METHOD

Bare-belt check: product holds only if $\mu \geq \tan(\theta)$; critical incline $\theta_c = \arctan(\mu)$.

Flap-load check: $F = m\ g\ (\sin(\theta) - \mu \cos(\theta))$, compared against the flap's limit before deflection under that load.

SIDE VIEW / INCLINED TRANSFER



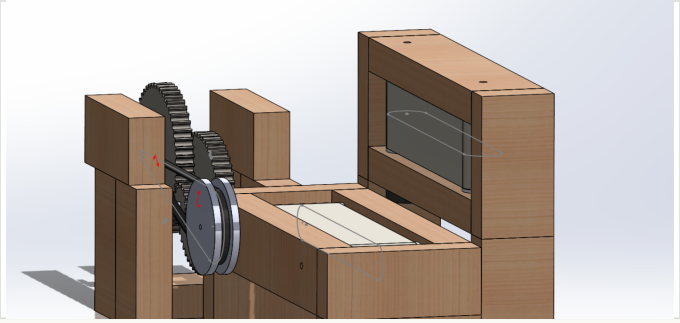
UHMW belt with retaining flaps. Schematic from my layout sketch; no dimensions are implied.

04 / UNIVERSITY OF GUELPH - ENGG 3280

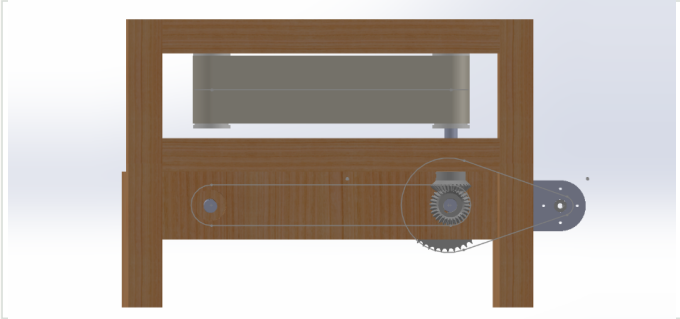
Selecting a drive for two belts

Third Year | September 2025 | Team of four
My role: CAD design, project coordination, drivetrain design and shaft selection

One 12 V DC motor drives a roadway belt under a stationary model car and a painted skyline belt through mechanical transmission. The motor is the only electrical component. The disassembled envelope is limited to 32 x 34.5 x 27.5 in; at least one shaft is required.



Initial CAD: multiple spur gears; component cost drove the redesign.



Final CAD: a single chain stage replaces the multi-gear arrangement.

OPTION	DECISION
Spur-gear train	Multiple gears increased component cost and drove the redesign.
Belt and pulley	Rejected over concern about slipping.
Chain and sprocket	Selected for positive engagement; noise accepted.

SELECTED CHAIN REDUCTION

8 T to 45 T | 5.625:1

About 889 rpm calculated at the measured 5,000 rpm no-load point; not a measured operating speed.

SELECTED MAIN SHAFT

20 mm

I selected an AISI 1030 cold-rolled steel main shaft.

A 30-tooth miter-gear pair transfers rotation to the background shaft at 1:1, so both belts share main-shaft speed.

ROADWAY BELT SPEED / CALCULATED NO-LOAD UPPER BOUND

Roadway only: v = pi x 0.060 m x 889/60 = **about 2.8 m/s**, not a measured speed. Shared main-shaft speed contributed to the fast scenery; this calculation does not give skyline-belt speed.

Other work completed by the team included detailed stress analysis, kinematics and motor characterization.

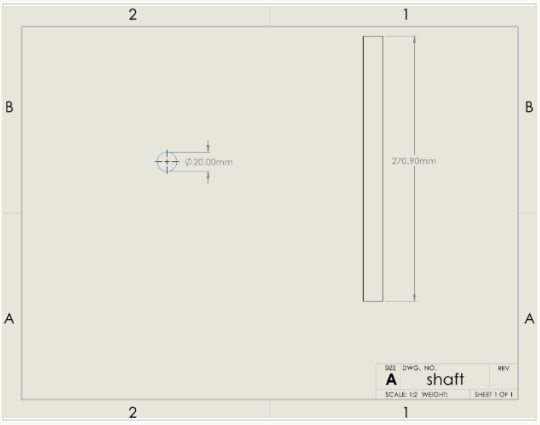
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Shaft interfaces and prototype testing

Third Year | September 2025 | Team of four
Component interfaces, running trials and design lessons



Finished mechanism: roadway belt, painted skyline belt and two model cars.



Main-shaft drawing: 20.00 mm diameter x 270.90 mm length.

SELECTED COMPONENTS AND INTERFACES

INTERFACE	SPECIFICATION
Shaft + bearings	20.00 x 270.90 mm shaft; 6204 bearings with 20 mm bores.
Miter gear	20 mm bore and 6 mm keyway.
Side panel	Two 47.00 mm bearing bores at 436.20 mm centres.
Drum + end cap	ABS drum: 60.00 mm OD, 50.00 mm bore, 102.87 mm face. End-cap flange: 75.00 mm.
Belt clearance	101.60 mm belt; 1.27 mm total side clearance.

PROTOTYPE TEST RECORD

CHECK	OBSERVED / RESPONSE
Chain drive	Ran the roadway belt with two model cars without slipping.
Miter-gear mesh	Ran the background belt.
Belt speed	Initial scenery speed was excessive; friction pads reduced the observed speed to approximately half of the initial value.
Guarding	Sheet-metal covers added.

PROPOSED, NOT IMPLEMENTED

Tensioners, CNC-cut frame members, interchangeable scenery and a greater reduction ratio.